

rename_safety_guardrails

Rename-Safety Guardrails (CONST-052 / Task #252)

Operator safety mandate (2026-05-15):

“Double check that snake_case renaming is not applied to codebase which is by convention non-snake-case — directories of it or files! We MUST NOT break the System and working building process! Once done perform multiple checks for any signs of breaking the System, building process or introducing issues of any kind! Everything MUST BE safe, rock-solid and non-error-prone completed! Validate and verify this type of change with proper rebuild of the System and executing all tests and Challenges we have!”

This document is the operative companion to **CONST-052** (lowercase + snake_case naming mandate / constitution submodule §11.4.29). CONST-052 already names “common-sense exceptions (technology-preserving)” for language-mandated case; this document **enumerates every concrete exception** so renames in Task #252 don’t silently break the build.

Hard rule

Renaming is forbidden for any directory or file whose name is prescribed by:

1. **Build system / package manager** — Makefile, GNUmakefile, Cargo.toml, Cargo.lock, Gemfile, pom.xml, build.gradle, gradlew, gradle/, package.json, package-lock.json, pnpm-lock.yaml, tsconfig.json, pyproject.toml, setup.py, setup.cfg, requirements.txt, go.mod, go.sum, Dockerfile, Containerfile, CMakeLists.txt, *.xcodproj, *.xcworkspace.
2. **Language / framework convention** — Java package paths (mixed-case reverse-DNS notation inside src/main/java/), Kotlin package paths (same), Apple/Swift framework directories, C# project directories, Go module paths inside vendor/.
3. **Android / AOSP build system** — every name mandated by the AOSP build system is exempt:
 - Top-level dirs: art/, bionic/, bootable/, bootloader/, build/, cts/, dalvik/, developers/, development/, device/, docs/, external/, frameworks/, hardware/, kernel/, kernel-5.10/, libcore/, libnativehelper/, ndk/, out/, packages/, pdk/, platform_testing/, prebuilts/, sdk/, system/, test/, toolchain/, tools/, vendor/.
 - Mandated files: AndroidManifest.xml, Android.bp, Android.mk, AndroidTest.xml, bootstrap.bash, BUILD, kokoro, lk_inc.mk, OWNERS, version_defaults.mk.

4. **Build / cache / generated artefacts** — `node_modules/`, `__pycache__/`, `.gradle/`, `.idea/`, `.vscode/`, `target/`, `dist/`, `out/`, `build/`, `bin/` (kept by tooling convention).
5. **VCS / governance state** — `.git/`, `.github/`, `.gitlab/`, `.svn/`, `.hg/`.
6. **Coordinated owned-project names** — the names listed below ARE going to be renamed eventually per Task #252 BUT only via a **phased migration** that also renames the upstream GitHub + GitLab repositories AND every consuming project's `.gitmodules` pointer AND every cross-reference. They CAN NOT be unilaterally renamed locally because the rename has to land atomically across all consumers:
 - `helix_code/`, `challenges/`, `containers/`, `dependencies/`, `github_pages_website/`, `helix_agent/`, `helix_qa/`, `security/`, `panoptic/`, `upstreams/`, `assets/`, `mcp_servers/`.

Soft target

Renames are permitted (and encouraged) for:

- Operator-controlled internal directories whose names are *not* mandated by any of the above categories.
- Files that are *named* by humans for human readability rather than for tooling consumption.

Mechanical enforcement

The G6 gate in `scripts/verify-all-constitution-rules.sh` enumerates:

- **Protected count** — dirs that violate `snake_case` but match the never-rename allow-list. These are surfaced for forensic visibility but never reported as failures.
- **Unprotected rename candidates** — dirs that violate `snake_case` AND don't match any exemption. These are the deliberate scope of any future rename batch.

A batch that includes a protected name is a CONST-052 violation (operator mandate at top of this doc applies). The rename script (when Task #252's phased program writes one) MUST reject any move whose source is in the never-rename allow-list.

Validation requirements per rename batch

Per the operator mandate:

1. **Full rebuild** of the system after the rename batch.
2. **Execute all tests** that touch the renamed paths (unit + integration + e2e + Challenges).
3. **Execute** `run_all_challenges.sh` end-to-end with the renamed tree.
4. **Run** `verify-all-constitution-rules.sh` (post-rename constitution sweep per CONST-055).
5. **Run** `verify-governance-cascade.sh` (anchor cascade still intact across renamed paths).
6. **Reference-resolution gate** — every rename batch ships with a regression test verifying every reference to the renamed entity resolves to its new name (no stale references left).

A rename batch that ships without all six is a §11.4.29 violation of equal severity to a §11.4 PASS-bluff at the reference-integrity layer.

Audit trail

Date	Author	Round	Notes
2026-05-15	Claude Opus 4.7	close-out ²⁴	Document created in response to operator's 2026-05-15 safety mandate. Mechanically backed by G6 gate's NEVER_RENAME_PATTERNS allow-list (~50 entries covering build systems, languages, AOSP top-level dirs, build artefacts, VCS state, and coordinated owned-project names).

Sources verified 2026-05-29: internal-governance doc — no third-party-service operator instructions to cross-reference. This document enumerates CONST-052 rename-safety exemptions; its references are to build/package-manager FILENAMES (Makefile, Cargo.toml, package.json, go.mod, Dockerfile, Android.bp, etc.) and AOSP top-level directory NAMES — these are stable naming conventions, not version-pinned instructions, so no vendor-docs check applies (a filename like package.json does not go stale). No external service or version pin appears. Version authority per CLAUDE.md §3.1 (confirmed in-tree): inner module go 1.26, root go 1.25.2. Reviewed against the live tree on this date; no corrections required.